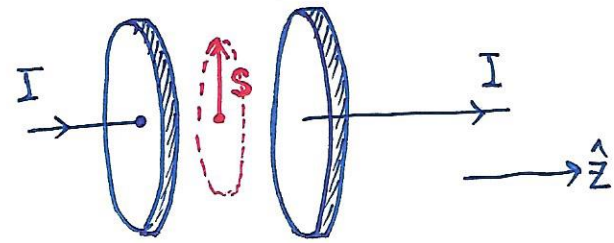
1.(a) Electric field in between the plates is given by

$$\vec{E}(t) = \frac{\sigma(t)}{\epsilon_0}(\hat{z}) = \frac{Q(t)}{\pi a^2 \epsilon_0}(\hat{z}) \quad \text{—(i)}$$

[direction of electric field is direction of current which is $\hat{z}$]

(b) Circle of radius 's' in the plane midway between the plates is shown - using the circle as the Amperian loop



$$\oint \vec{B}\cdot d\vec{l} = \mu_0 (I_d)_{enc} \quad \text{—(ii)}$$

$(I_d)_{enc}$ = displacement current

$$\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \text{ : displacement current density}$$

$$= \frac{I(t)}{\pi a^2}(\hat{z}) \text{ — from (i)}$$

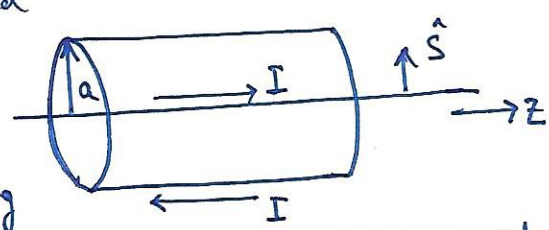
$$\therefore I_d = \oint_S \vec{J}_d \cdot d\vec{a} = \frac{I(t)}{\pi a^2} \times \pi s^2 = I(t)\frac{s^2}{a^2}.$$

This is the enclosed displacement current.

Applying (ii) $B \cdot 2\pi s = \mu_0 I(t) \frac{s^2}{a^2} \Rightarrow \boxed{\vec{B} = \frac{\mu_0 I(t) s}{2\pi a^2}\hat{\varphi}}$

[$\vec{B}$ along $\hat{\varphi}$ because $\vec{J}_d$ is longitudinal ($\hat{z}$)]

2. In Q5 of Tutorial 14, we had shown that if the current flowing through the cylindrical tube of radius a and returning back along the outer surface (as shown) varied sinusoidally $I(t) = I_0 \cos\omega t$, the induced electric field $\vec{E}(s,t)$ was given by $\vec{E}(s,t) = \frac{\mu_0 I_0 \omega}{2\pi} \sin(\omega t) \ln\left(\frac{a}{s}\right)\hat{z}$. Note that the induced



electric field points along z-direction.

(a) $\vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\mu_0 \epsilon_0 I_0 \omega^2}{2\pi} \cos(\omega t) \ln\left(\frac{a}{s}\right) \hat{z}$

$\cancel{\mu_0 \epsilon_0 = \frac{1}{c^2}}$; $I = I_0 \cos \omega t$

$$\vec{J}_d = \frac{I \omega^2}{2\pi} \ln\left(\frac{a}{s}\right) \hat{z} \times \mu_0 \epsilon_0$$

(b) Use $I_d = \oint \vec{J}_d \cdot \vec{da}$ where $\vec{da}$ = area element for a surface of constant z

$$I_d = \int_0^a \frac{I \omega^2}{2\pi} \ln\left(\frac{a}{s}\right) s\, ds \int_0^{2\pi} d\varphi \quad (\times \mu_0 \epsilon_0)$$

$$\boxed{\begin{aligned} \vec{da} &= dl_s\, dl_\varphi (\hat{z}) \\ &= ds\,(s\, d\varphi)\hat{z} = s\, ds\, d\varphi\, (\hat{z}) \end{aligned}}$$

$$= \frac{\mu_0 \epsilon_0 I \omega^2}{\cancel{2\pi}} \int_0^a s \ln\left(\frac{a}{s}\right) ds \times \cancel{2\pi} = \mu_0 \epsilon_0 I \omega^2 \int_0^a (s \ln a - s \ln s)\, ds$$

$$= \mu_0 \epsilon_0 I \omega^2 \left[(\cancel{\ln a}) \frac{a^2}{2} - (\cancel{\ln a}) \frac{a^2}{2} + \frac{a^2}{4} \right]$$

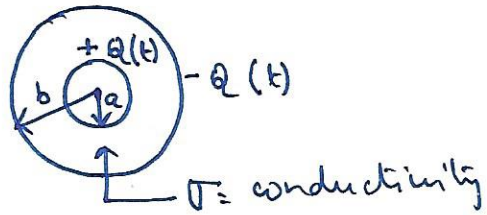
$$= \boxed{\mu_0 \epsilon_0 I \omega^2 \frac{a^2}{4}}$$

[Integration by parts

$\int u v\, dx = u \int v\, dx - \int \left[\frac{du}{dx} \int v\, dx \right] dx$

Take $u = \ln s$]

3. $\vec{J} = \sigma \vec{E}$



(a) Since $\vec{E} = \frac{Q(t)}{4\pi\epsilon_0 r^2} (\hat{r})$ is radial, the current density $\vec{J}$ would also be radial.

$$\vec{J} = \frac{\sigma\, Q(t)}{4\pi\epsilon_0 r^2} \hat{r} \quad \text{(conduction current density)}$$